Mask-Guided Correlation Learning for Few-Shot Segmentation in Remote Sensing Imagery

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***Abstract*— Few-shot segmentation aims to segment specific objects in a query image based on a few densely annotated images and has been extensively studied in recent years. In remote sensing, image segmentation faces challenges such as less training data, large intraclass diversity, and low foreground–background contrast. In this work, we propose a novel few-shot segmentation method in remote sensing imagery based on mask-guided correlation learning (MGCL) to alleviate the above challenges. In our MGCL, a novel mask-guided feature enhancement (MGFE) module is proposed, which makes features have intramask consistency by leveraging oversegmented masks. In order to enhance the contrast between foreground and background, a novel foreground–background correlation (FBC) module is proposed, which enhances background correlation representation by learning foreground correlation and background correlation separately. Furthermore, a novel mask-guided correlation decoder (MGCD) module is proposed to guide the decoder to focus on the consistency within the mask, thereby learning how to segment complete objects and improving segmentation accuracy. Sufficient experiments on the iSAID-5^i and DLRSD-5^i datasets show that our MGCL outperforms all comparative methods. In particular, in the one-shot setting of the iSAID-5^i dataset, we achieve an mIoU of 39.92 based on ResNet50, which is an improvement of 4.25 over the state-of-the-art (SOAT) method. The visualization of features before and after the MGFE module further concretely demonstrates the motivation and advantages of our MGCL. The code is available at https://github.com/LiShuo1001/MGCL.**

***Index Terms*— Correlation learning, few-shot learning, few-shot segmentation, segment anything, semantic segmentation.**

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I. Introduction

SEMANTIC segmentation aims to segment objects with semantic categories, and deep learning-based methods have made breakthrough progress in natural images [1], [2], [3], [4]. With the development of remote sensing technologies, semantic segmentation has many important application scenarios in remote sensing [5], [6], [7], such as agricultural management [8], [9], [10], forest management [11], [12], and disaster detection [13], [14], [15]. However, deep learning-based semantic segmentation models heavily rely on large-scale annotated data. This means that obtaining numerous high-quality dense annotations consumes a lot of manpower and material resources [16]. Furthermore, it is often difficult to obtain sufficient natural images or remote sensing images, resulting in some categories not having data that can meet minimum training requirements. Unfortunately, we sometimes encounter the need to segment unseen classes [17], [18]. The amount of data for these unseen classes is too small to train suitable models, and existing models trained on a large amount of labeled data for other classes often fail to generalize to unseen classes. In order to alleviate the above problems, few-shot segmentation is proposed, which aims to train a model that can predict segmentation maps for query images based on a few annotated support images [19], [20], [21], [22].

Recently, some researchers have applied few-shot segmentation to the field of remote sensing by combining the characteristics of this field [17], [23], [24], [25]. As shown in Fig. 1, remote sensing images usually have challenges and difficulties, such as large intraclass diversity, low foreground–background contrast, and large-scale changes. Aiming at a large variance of scale, Yao et al. [23] propose a scale-aware network to perform detailed matching with multiple prototypes. Targeting irrelevant distractors, Lang et al. [24] propose a progressive parsing module and a commonality distillation module to remedy the frequent occurrence of irrelevant distractors. To account for the huge difference in appearance and scales, Wang et al. [17] enhance the generalization ability by considering the self-correlation and cross correlation between the support and query images. Although these methods have made progress on the challenges and difficulties of remote sensing images, they do not explicitly model large intraclass diversity and low foreground–background contrast, resulting in

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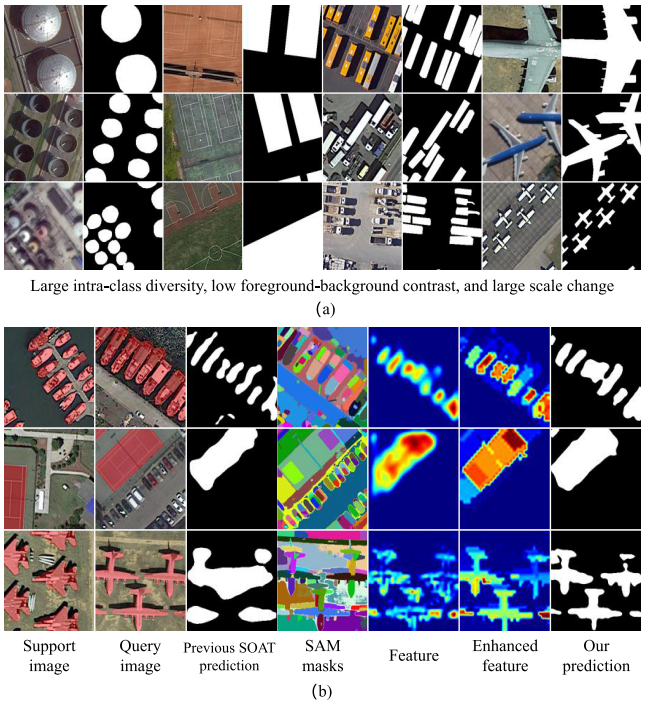


Fig. 1. (a) In the first half, challenges such as large intraclass diversity, low foreground–background contrast, and large-scale changes that are prevalent in remote sensing imagery are demonstrated. (b) In the second half, the motivation of our work is presented: guiding feature enhancement through oversegmented masks to achieve the goal of segmenting complete objects. (a) Challenges in remote sensing imagery. (b) Motivation of our method.

predicted segmentation maps with incomplete objects, inaccurate boundaries, and so on, as shown in Fig. 1 for the prediction results of previous state-of-the-art (SOAT) methods.

With the growth of natural image data scale and the development of Transformer [26] technology, the segmentation anything model (SAM) aimed at segmenting everything has been proposed [27]. In SAM, all object masks in the image are predicted interactively, and these objects represented by the masks usually appear to be oversegmented, as shown in Fig. 1. Although the objects are oversegmented, these masks have their own specific semantic information and also indicate the boundaries of the objects. Therefore, *this work will investigate how to benefit from these oversegmented masks* to alleviate challenges and difficulties such as large intraclass diversity and low foreground–background contrast, thereby achieving segmentation of complete objects and accurate boundaries based on annotated supporting images.

To benefit from oversegmented masks, we propose a mask-guided correlation learning (MGCL) method for few-shot segmentation in remote sensing imagery, as illustrated in Fig. 2. In our MGCL, by using oversegmented masks to guide how to enhance features, we propose a novel mask-guided feature enhancement (MGFE) module that makes the enhanced features have intramask feature consistency, thereby reducing intraobject interference and providing high-quality features for segmenting complete objects. By learning foreground correlation and background correlation separately, we propose a novel foreground–background correlation (FBC) module that enhances the representation of background correlation and reduces the impact of low foreground–background contrast. In the decoding of correlation learning, we propose a novel mask-guided correlation decoder (MGCD) module that guides the decoder to pay attention to the consistency within the mask and learn how to segment complete objects, thereby improving the accuracy of segmentation. Overall, our proposed method achieves the best results on two common datasets and establishes a large performance gap with previous methods. As shown in Fig. 1, the visualization of features before and after MGFE and the prediction results further demonstrates the motivation and advantages of our proposed method.

Compared to previous methods [17], [23], through the MGFE and FBC modules, our proposed MGCL achieves intramask feature consistency and foreground–background discriminability. These two points of our method are very beneficial for improving the completeness of segmented objects and suppressing irrelevant background interference. To summarize, the following are our main contributions.

1) An MGCL method is proposed for few-shot segmentation in remote sensing, which achieves intramask feature consistency and models the correlation of foreground and background separately.
2) An MGFE module based on oversegmented masks is designed to achieve intramask feature consistency and reduce intraobject interference.
3) To improve the discriminability between background and foreground, an FBC module is constructed to significantly enhance the representation of background correlation.
4) Experimental results show that our method achieves SOAT results on both iSAID-5^i and DLRSD-5^i datasets. The visualization also shows that our method achieves MGFE and segments more complete objects with more accurate boundaries.

Our work is organized as follows. After introducing the related work in Section II, the methods are introduced in Section III. Then, the results are reported in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

In this section, related works in three topics are mainly introduced: few-shot segmentation (see Section II-A), few-shot segmentation in remote sensing imagery (see Section II-B), and correlation learning for few-shot segmentation (see Section II-C).

A. Few-Shot Segmentation

Few-shot segmentation aims to segment new objects in a query image from one or a few annotated support images [18] and often requires learning how to segment unseen classes [28]. Shaban et al. [18] extend low-shot learning techniques for image classification to dense semantic segmentation, pioneering the field of few-shot segmentation.

In recent years, many methods focusing on few-shot segmentation have been proposed by the authors [29], [30], [31], [32], [33], [34], [35], [36], [37], [38], which can be roughly

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divided into three categories: conditional network-based methods [29], [30], latent space optimization-based methods [35], [36], [37], and [38], and prototypical learning-based methods [31], [32], [33]. **Conditional network-based methods** usually contain parallel conditional branch and segmentation branch [18], [29], [30]. The conditional branch takes the annotated support images as input and generates a parameter set, and the segmentation branch takes the query image and the parameter set as input and, finally, generates a prediction segmentation mask [29]. **Latent space optimization-based methods** can be further refined into methods based on generative models [35], [35], [36], [37], [38], [39], contrastive learning [35], [39], [40], and variational autoencoders [38], [41]. Generative models often have a complex training process, and the trained model can be used as a feature extractor [35], [36] or generate new synthetic data [37]. Saha et al. [35] predict segmentation masks using features extracted from networks trained in contrastive learning. Sun et al. [41] enhance the model by incorporating a variational attention mechanism to highlight the foreground of the query image. **Prototypical learning-based methods** typically perform metric learning on features of the support and query images [31], [32], [33], [42], [43]. These methods are usually based on the average of the features related to the same class to represent the prototype features [44], [45], [46], [47], [48]. Since there is a large difference between the foreground object and the background in natural images, prototypical learning-based methods usually only focus on the prototype of the foreground object and ignore the prototype of the background [31], [48]. In our method, in order to increase the prediction ability of the model for low foreground–background contrast images in remote sensing, we propose an FBC module that establishes feature prototypes for foreground and background, respectively.

B. Few-Shot Segmentation in Remote Sensing Imagery

In recent years, due to the advantages of few-shot segmentation, many few-shot segmentation methods that combine remote sensing characteristics have been proposed in remote sensing imagery [17], [23], [24], [49], [50], [51], [52].

A common strategy to improve segmentation accuracy is to exploit additional information related to remote sensing images [51], [52]. For example, Lang et al. [52] propose a few-shot segmentation framework with global rectification and decoupled registration, which utilizes a series of dynamically updated global prototypes to provide auxiliary nontarget segmentation cues. In this framework, global prototypes increase the negative impact of scale changes and low foreground–background contrast; Wu et al. [51] combine prototype information with global edge information to achieve more accurate prototype-matching results, which not only increases the computational cost and model complexity but also results in lower matching accuracy based on edge information due to the large intraclass differences in remote sensing images.

Incorporating remote sensing characteristics is also a common important improvement strategy [23], [24]. For example, by combining the characteristics of large-scale changes, Yao et al. [23] propose a scale-aware network to perform detailed matching with multiple prototypes, which alleviates the challenge of scale changes but fails to handle the confusion caused by background and foreground similarities. To reduce the interference of low foreground–background contrast, Lang et al. [24] propose a progressive parsing module and a commonality distillation module to remedy the frequent occurrence of irrelevant terms, which alleviates the challenge of low foreground–background contrast but lacks a mechanism to cope with scale changes. Wang et al. [17] enhance the generalization ability by considering the self-correlation and cross correlation between the support and query images, which overemphasizes the correlation of the foreground and lacks the correlation modeling of the background. This idea of computing correlation provides a key inspiration for our FBC module.

In order to obtain complete segmentation objects, we utilize class-agnostic masks as additional information and propose an MGFE module to alleviate the phenomenon of large intraclass diversity. Furthermore, we model the correlation of foreground and background separately and propose an FBC module to reduce the impact of low foreground–background contrast on model performance.

C. Correlation Learning for Few-Shot Segmentation

Recently, dense semantic correlation learning has been studied in low-level vision [53]. The goal of dense semantic correlation learning is to find reliable correspondence between images despite the challenge of large intraclass variation [54], [55]. Min et al. [20] propose a hypercorrelation squeeze network utilizing multilevel feature correlation and efficient 4-D convolutions, which extracts features from different levels of intermediate convolutional layers and builds a collection of 4-D correlation tensors. Zheng et al. [56] introduce a quaternion perspective on correlation learning and propose a quaternion-valued correlation learning network designed to reduce the computational burden of high-dimensional correlation tensors. Based on the work of Min et al. [20], Wang et al. [17] introduce self-correlation learning of query images to enhance the generalization ability on the basis of correlation learning between the support and query images. The above correlation learning only focuses on and learns the correlation of the foreground while ignoring the interference caused by the background. Combined with the characteristics of low foreground–background contrast in remote sensing images, we propose to learn correlations of the foreground and background separately based on correlation learning and jointly encode the FBC features in a 4-D convolutional pyramid encoder. In this way, our method improves prediction performance on large intraclass diversity and low foreground–background contrast images.

III. Our Approach

A. Task Description

Few-shot segmentation aims to segment objects in the query image, which have the same semantic categories as support images based on a few untrained annotated support images.

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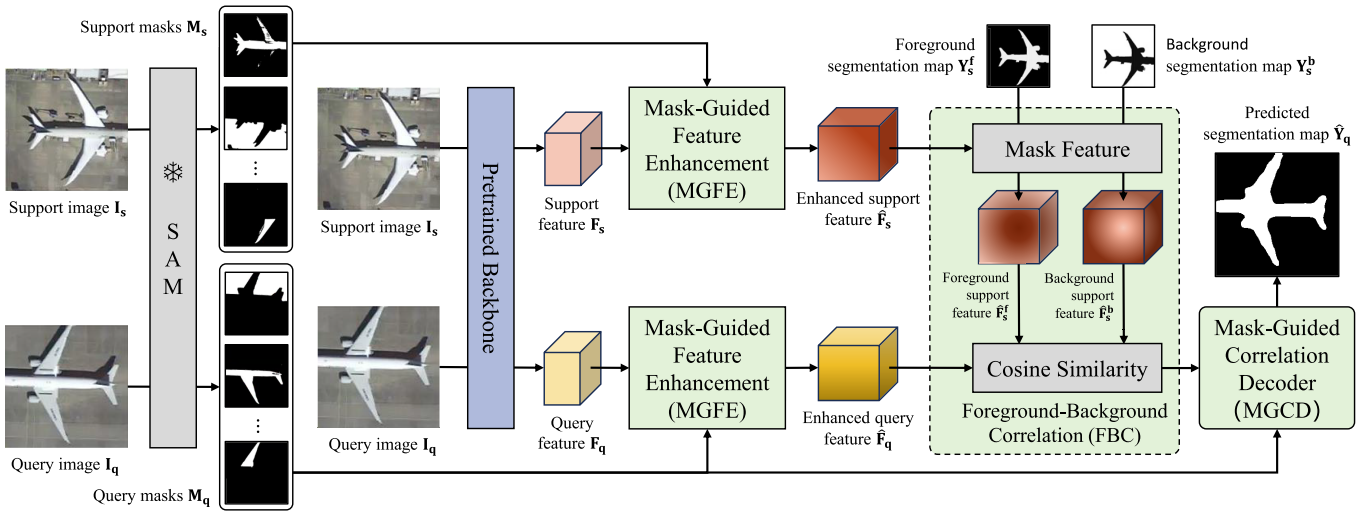


Fig. 2. Overall pipeline of our MGCL. Our method includes a frozen SAM, a pretrained backbone, two MGFE modules (see Section III-C), an FBC module (see Section III-D), and an MGCD module (see Section III-E). Among the above modules, MGFE, FBC, and MGCD are the main contributions and novelties of our method. MGFE aims to utilize oversegmented masks ($\mathbf{M_s}$ and $\mathbf{M_q}$) to enhance the intramask consistency of the extracted features ($\mathbf{F_s}$ and $\mathbf{F_q}$), as detailed in Fig. 3. FBC consists of the mask feature module and the cosine similarity module. FBC consists of a mask feature module and a cosine similarity module, which is designed to model the correlation of the foreground and the background respectively, where the cosine similarity module is shown in Fig. 4. MGCD aims to predict the final segmentation map based on the L stage foreground–background correlations, as shown in Fig. 5.

The semantic categories in a dataset are usually divided into nonoverlapping base classes $\mathcal{C}_{\text{base}}$ and unseen classes $\mathcal{C}_{\text{unseen}}$, that is, $\mathcal{C}_{\text{base}} \cap \mathcal{C}_{\text{unseen}} = \emptyset$. In order to have the ability to learn how to segment based on a few annotated support images, first, train a segmentation model f on a training set $\mathcal{D}_{\text{train}}$ constructed from the data of base classes, and then evaluate the learned ability of the segmentation model f on a test set $\mathcal{D}_{\text{test}}$ constructed from the data of unseen classes.

As with previous methods [17], [23], we use the K-shot setting to randomly sample annotated images to form a large number of episodes. Each episode contains $K+1$ image-label pairs and forms a support set $\mathcal{S} = \{(\mathbf{I_s^k}, \mathbf{Y_s^k})\}_{k=1}^{K}$ and a query set $\mathcal{Q} = \{(\mathbf{I_q}, \mathbf{Y_q})\}$, where the foreground objects indicated in $\mathbf{Y_s^k}$ and $\mathbf{Y_q}$ have the same semantic category. Both the training set $\mathcal{D}_{\text{train}}$ and the test set $\mathcal{D}_{\text{test}}$ are composed of numerous episodes. During the training phase, the segmentation model f takes the support set $\mathcal{S}$ and the query image $\mathbf{I_q}$ from the training set $\mathcal{D}_{\text{train}}$ as inputs to predict a segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$ and calculates the cross entropy loss between $\mathbf{Y_q}$ and $\hat{\mathbf{Y}}_\mathbf{q}$ to update the weight θ

$$\hat{\mathbf{Y}}_\mathbf{q} = f_\theta\left(\{(\mathbf{I_s^k}, \mathbf{Y_s^k})\}_{k=1}^{K}, \mathbf{I_q}\right). \tag{1}$$

During the test phase, the segmentation model f takes the support set $\mathcal{S}$ and the query image $\mathbf{I_q}$ from the test set $\mathcal{D}_{\text{test}}$ as inputs to inference a final segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$.

B. Method Overview

Our method utilizes oversegmented masks to guide correlation learning in two aspects: 1) guide the support feature and the query feature to enhance their intramask consistency and 2) guide the decoder to improve the intramask consistency of the predicted segmentation map. The overall architecture of our proposed method is shown in Fig. 2. The novelty of our method is mainly reflected in MGFE, FBC, and MGCD.

In our method, first, given the support image $\mathbf{I_s}$ and the query image $\mathbf{I_q}$, the oversegmented support masks $\mathbf{M_s}$ and the oversegmented query masks $\mathbf{M_q}$ are predicted by the frozen SAM. For a fair comparison with related methods [17], [49], [52], we extract the support feature $\mathbf{F_s}$ of the support image and the query feature $\mathbf{F_q}$ of the query image from the pretrained backbone, such as VGG16 [57] and ResNet50 [58]. Then, in our MGFE module, the oversegmented masks are used to guide how to enhance the features to obtain consistent features within each mask. Specifically, $\mathbf{M_s}$ is used to enhance the support feature $\mathbf{F_s}$ to obtain the enhanced support feature $\hat{\mathbf{F}}_\mathbf{s}$, and $\mathbf{M_q}$ is used to enhance the query feature $\mathbf{F_q}$ to obtain the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$. Then, in our FBC module, the correlation between each pixel in the enhanced support feature $\hat{\mathbf{F}}_\mathbf{s}$ and the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$ is calculated. In order to fully mine the correlation, we split the segmentation map $\mathbf{Y_s}$ of the support image into a foreground segmentation map $\mathbf{Y_s^f}$ and a background segmentation map $\mathbf{Y_s^b}$. Through the mask feature module, the foreground support feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}$ and the background support feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}$ are obtained by masking the enhanced support feature $\hat{\mathbf{F}}_\mathbf{s}$ using $\mathbf{Y_s^f}$ and $\mathbf{Y_s^b}$, respectively. Through the cosine similarity module, the 4-D correlations are quantified using the cosine similarity between each pixel in $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}$ and all pixels in $\hat{\mathbf{F}}_\mathbf{q}$. Likewise, the 4-D correlations between each pixel in $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}$ and all pixels in $\hat{\mathbf{F}}_\mathbf{q}$ are also calculated. Finally, in our MGCD module, the 4-D correlations are decoded to predict the final segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$. In our MGCD module, the oversegmented query masks $\mathbf{M_q}$ are used to guide the decoding process to improve the consistency within each mask in the final segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$.

In the following subsections, we will introduce our proposed MGFE, FBC, and MGCD modules in detail respectively.

C. Mask-Guided Feature Enhancement

Pixels indicated in an oversegmented mask generally have similar appearance or semantics. These pixels should have similar features to exhibit this similar appearance or semantics.

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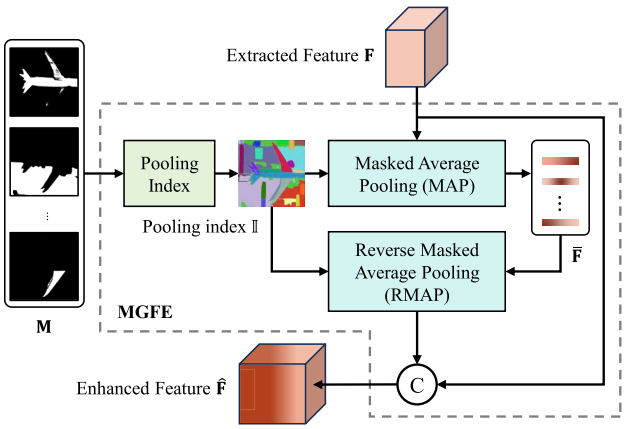


Fig. 3. Detailed illustration of our MGFE module. This module takes masks $\mathbf{M}$ (the support masks $\mathbf{M_s}$ or the query masks $\mathbf{M_q}$) and extracted feature $\mathbf{F}$ (the support feature $\mathbf{F_s}$ or the query feature $\mathbf{F_q}$) as input and outputs enhanced feature $\hat{\mathbf{F}}$ (the enhanced support feature $\hat{\mathbf{F}}_\mathbf{s}$ or the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$). The circled C is the concatenation operation.

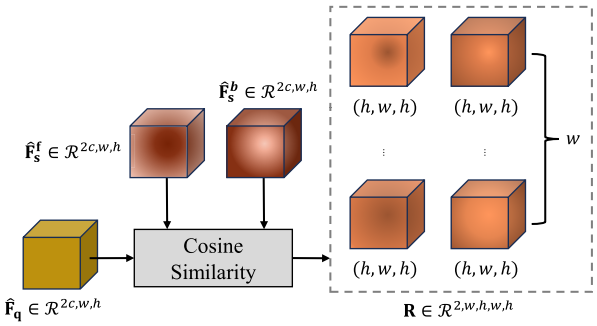


Fig. 4. Inputs and outputs of the cosine similarity module.

In practice, features extracted through the backbone network sometimes do not have this similarity. Therefore, as shown in Fig. 3, we propose an MGFE module to utilize oversegmented masks to enhance the intramask consistency of the extracted features.

In our proposed MGFE module, each mask is first assigned an index for pooling the extracted features. Since there may be overlap between different masks, we assign different index values in sequence according to the area size of the mask. Specifically, after masks are generated, we sort them according to the area size of the mask and assign an index to each mask, and for batch training, we only keep the top m masks. That is, if the number of masks is more than m, the mask with a smaller mask area will be removed, and if the number of masks is less than m, an empty mask will be filled. In Fig. 3, different colors in the color map after pooling index represent different indexes, and the same color represents the same index. The above process can be expressed as

$$\mathbb{I} = \text{PoolingIndex}(\mathbf{M}) \tag{2}$$

where $\mathbf{M}$ is the support masks $\mathbf{M_s}$ or the query masks $\mathbf{M_q}$. $\mathbb{I}$ represents the polling index $\mathbb{I}_\mathbf{s}$ of the support masks $\mathbf{M_s}$ or the polling index $\mathbb{I}_\mathbf{q}$ of the query masks $\mathbf{M_q}$.

Then, according to the pooling index $\mathbb{I}$, the extracted feature $\hat{\mathbf{F}}$ is average pooled through a masked average pooling (MAP) module. The MAP module can be expressed as

$$\bar{\mathbf{F}} = \text{MAP}(\mathbf{F}, \mathbb{I}) \tag{3}$$

where $\mathbf{F} \in \mathcal{R}^{c\times w\times h}$ is the support feature $\mathbf{F_s}$ or the query feature $\mathbf{F_q}$, and c, w, and h are the dimensions, width, and height of the feature, respectively. The details of the MAP module are given as follows:

$$\bar{\mathbf{F}} = \{\bar{\mathbf{F}}_i\}_{i=1}^m, \quad \bar{\mathbf{F}}_i = \frac{1}{||\mathbb{I}_i||}\sum \mathbb{I}_i * \mathbf{F} \tag{4}$$

where m is the number of masks, m is also the number of polling indexes, $\mathbb{I}_i$ is the one-hot index of the ith mask, $||\mathbb{I}_i||$ represents the number of pixels of $\mathbb{I}_i$, $\bar{\mathbf{F}}_i \in \mathcal{R}^c$ represents the average of the features indicated by the index of the ith mask, and $\bar{\mathbf{F}} \in \mathcal{R}^{m\times c}$ is the feature after average pooling.

Then, in order to obtain consistent features within the mask, we propose a reverse masked average pooling (RMAP) module to convert the 2-D feature $\bar{\mathbf{F}}$ into a 3-D feature $\hat{\mathbf{F}} \in \mathcal{R}^{c\times w\times h}$. The above process can be expressed as

$$\hat{\mathbf{F}} = \text{ReverseMAP}(\bar{\mathbf{F}}, \mathbb{I}) \;||\; \mathbf{F} \tag{5}$$

where $||$ is the concatenation operation and $\hat{\mathbf{F}} \in \mathcal{R}^{2c\times w\times h}$. The enhanced feature $\hat{\mathbf{F}}$ can be the enhanced support feature $\hat{\mathbf{F}}_\mathbf{s}$ or the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$.

The function of the MGFE module is to assimilate the features within a mask, that is, to make the features within a mask tend to be the same, which maintains the feature consistency of pixels with similar appearance or semantics, providing enhanced features for the decoder to predict more accurate and precise segmentation map results.

D. Foreground–Background Correlation

Few-shot segmentation aims to segment out foreground objects in the query image, where these foreground objects are exemplified by the support images and their segmentation maps. The images consist of foreground objects and background. Compared with natural images, remote sensing images usually have low foreground–background contrast. Therefore, modeling the similarity between the background pixels of the support and query images will help increase the distance between foreground and background in the feature space, thereby improving the performance of segmentation maps.

In order to achieve the above purpose, we propose an FBC module to model the correlation between the query images and the foreground of the support images and the correlation between the query images and the background of the support images, respectively. First, as shown in Fig. 2, the ground-truth segmentation map $\mathbf{Y_s}$ of the support image is divided into a foreground segmentation map $\mathbf{Y_s^f}$ and a background segmentation map $\mathbf{Y_s^b}$. Then, through a mask feature module, the foreground feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}$ and the background feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}$ of the support images are obtained

$$\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f} = \mathbf{Y}_\mathbf{s}^\mathbf{f} \odot \hat{\mathbf{F}}_\mathbf{s}, \quad \hat{\mathbf{F}}_\mathbf{s}^\mathbf{b} = \mathbf{Y}_\mathbf{s}^\mathbf{b} \odot \hat{\mathbf{F}}_\mathbf{s}. \tag{6}$$

where $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f} \in \mathcal{R}^{2c\times w\times h}$, $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b} \in \mathcal{R}^{2c\times w\times h}$, and $\odot$ is the Hadamard product. Finally, as shown in Fig. 4, correlations are calculated through a cosine similarity module. Specifically, the pairwise

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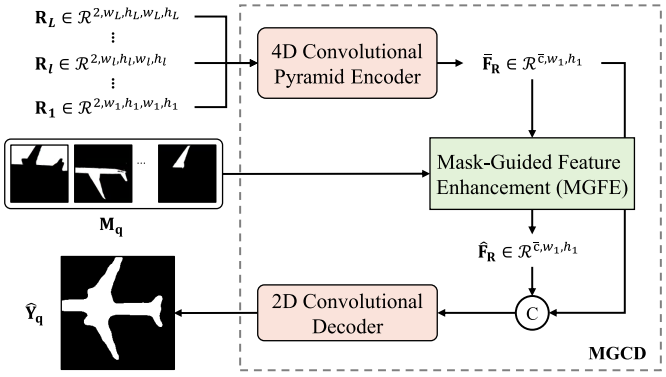


Fig. 5. Illustration of the MGCD module, which contains an MGFE module before 2-D convolutional decoder.

correlations of pixels between the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$ and the foreground feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}$, and the enhanced query feature $\hat{\mathbf{F}}_\mathbf{q}$ and the background feature $\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}$ are calculated, respectively,

$$\mathbf{R}^\mathbf{f}(i_\mathrm{q}, j_\mathrm{q}, i_\mathrm{f}, j_\mathrm{f}) = \sigma\left(\frac{\hat{\mathbf{F}}_\mathbf{q}(i_\mathrm{q}, j_\mathrm{q}) \cdot \hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}(i_\mathrm{f}, j_\mathrm{f})}{||\hat{\mathbf{F}}_\mathbf{q}(i_\mathrm{q}, j_\mathrm{q})||\ ||\hat{\mathbf{F}}_\mathbf{s}^\mathbf{f}(i_\mathrm{f}, j_\mathrm{f})||}\right)$$
$$\mathbf{R}^\mathbf{b}(i_\mathrm{q}, j_\mathrm{q}, i_\mathrm{b}, j_\mathrm{b}) = \sigma\left(\frac{\hat{\mathbf{F}}_\mathbf{q}(i_\mathrm{q}, j_\mathrm{q}) \cdot \hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}(i_\mathrm{b}, j_\mathrm{b})}{||\hat{\mathbf{F}}_\mathbf{q}(i_\mathrm{q}, j_\mathrm{q})||\ ||\hat{\mathbf{F}}_\mathbf{s}^\mathbf{b}(i_\mathrm{b}, j_\mathrm{b})||}\right) \quad (7)$$

where $(i_\mathrm{q}, j_\mathrm{q})$, $(i_\mathrm{f}, j_\mathrm{f})$, and $(i_\mathrm{b}, j_\mathrm{b})$ denote 2-D spatial positions of features, σ is the ReLU activation function, $||\cdot||$ represents the magnitude of the input, and $\mathbf{R}^\mathbf{f} \in \mathcal{R}^{w\times h\times w\times h}$ and $\mathbf{R}^\mathbf{b} \in \mathcal{R}^{w\times h\times w\times h}$ are the foreground and background correlations, respectively. By merging $\mathbf{R}^\mathbf{f}$ and $\mathbf{R}^\mathbf{b}$ together, we get the foreground–background correlation $\mathbf{R}$

$$\mathbf{R} = \mathbf{R}^\mathbf{f}\ ||\ \mathbf{R}^\mathbf{b} \quad (8)$$

where $\mathbf{R} \in \mathcal{R}^{2\times w\times h\times w\times h}$ and $||$ is the concatenation operation. Subsequently, we will decode the segmentation map of the query image from the correlation $\mathbf{R}$.

Since the scale of objects in remote sensing images varies greatly, similar to SCCNet [17] and HSNet [20], we adopt a feature pyramid network (FPN) [59] structure to improve the performance of our proposed method. Specifically, given the support image $\mathbf{I_s}$ and the query image $\mathbf{I_q}$, we extract L stage feature maps from the pretrained backbone, denoted as $\{\mathbf{F_s}^l, \mathbf{F_q}^l\}_{l=1}^L$, where $\mathbf{F_s}^l$ and $\mathbf{F_q}^l$ represent the support features and the query features at the lth stage, respectively. Based on the support features $\mathbf{F_s}^l$, the support masks $\mathbf{M_s}$, the query features $\mathbf{F_q}^l$, and the query masks $\mathbf{M_q}$, we can get the L stage foreground–background correlations $\{\mathbf{R}_l\}_{l=1}^L$, where $\mathbf{R}_l \in \mathcal{R}^{2\times w_l\times h_l\times w_l\times h_l}$.

E. Mask-Guided Correlation Decoder

In this subsection, as shown in Fig. 5, we propose an MGCD module to predict the final segmentation map based on the L stage foreground–background correlations $\{\mathbf{R}_l\}_{l=1}^L$.

Following previous works [17], [20], we encode $\{\mathbf{R}_l\}_{l=1}^L$ through a 4-D convolutional pyramid encoder module to obtain the correlation feature $\bar{\mathbf{F}}_\mathbf{R}$. Specifically, first, the 4-D convolutional pyramid encoder module utilizes L 4-D convolution blocks [20] to capture the preliminary correlation features of 4-D correlations

$$\hat{\mathbf{F}}_l = \mathrm{ConvB}_{4\mathrm{D}}^l(\mathbf{R}_l) \quad (9)$$

where $\mathrm{ConvB}_{4\mathrm{D}}^l$ is the lth 4-D convolution block, which consists of three sequences of center-pivot 4-D convolution [20], group normalization [60], and ReLU activation function. Then, the correlation feature $\hat{\mathbf{F}}_l$ of the previous $(l-1)$th layer is mixed with the correlation feature $\hat{\mathbf{F}}_{l-1}$ of the current lth layer after a $2\times$ upsampling layer

$$\bar{\mathbf{F}}_{l-1} = \mathrm{ConvB}_{4\mathrm{D}}^{l\to l-1}\left(\mathrm{UpS}_{2\times}(\hat{\mathbf{F}}_l) + \hat{\mathbf{F}}_{l-1}\right) \quad (10)$$

where $\mathrm{UpS}_{2\times}$ is the $2\times$ upsampling layer via bilinear interpolation and $\mathrm{ConvB}_{4\mathrm{D}}^{l\to l-1}$ is also a 4-D convolution block with the same layer structure as $\mathrm{ConvB}_{4\mathrm{D}}^l$. Finally, by performing average pooling on the last two spatial dimensions of $\bar{\mathbf{F}}_1$, the correlation feature $\bar{\mathbf{F}}_\mathbf{R} \in \mathcal{R}^{\bar{c}\times w_1\times h_1}$ is obtained, which converts a 4-D feature into a 2-D feature.

In order to make the correlation feature consistent within the mask, we enhance the correlation feature $\bar{\mathbf{F}}_\mathbf{R}$ through our proposed MGFE module to obtain the enhanced correlation feature $\hat{\mathbf{F}}_\mathbf{R}$, as shown in Fig. 5. By concatenating $\bar{\mathbf{F}}_\mathbf{R}$ and $\hat{\mathbf{F}}_\mathbf{R}$ together, we get the final correlation feature $\mathbf{F}_\mathbf{R}$

$$\mathbf{F}_\mathbf{R} = \hat{\mathbf{F}}_\mathbf{R}\ ||\ \bar{\mathbf{F}}_\mathbf{R}, \quad (11)$$

where $\mathbf{F}_\mathbf{R} \in \mathcal{R}^{2\bar{c}\times w_1\times h_1}$ and $||$ is the concatenation operation.

In our MGCD module, the 2-D convolutional decoder contains two convolutional blocks ($\mathrm{ConvB}_{2\mathrm{D}}^1$ and $\mathrm{ConvB}_{2\mathrm{D}}^2$) and a $2\times$ upsampling layer, in which each convolutional block consists of two 2-D convolution layers [17], [20]. Formally, the 2-D convolutional decoder can be expressed as

$$\hat{\mathbf{Y}}_\mathbf{q} = \sigma\left(\mathrm{ConvB}_{2\mathrm{D}}^2\left(\mathrm{UpS}_{2\times}\left(\mathrm{ConvB}_{2\mathrm{D}}^1(\mathbf{F}_\mathbf{R})\right)\right)\right) \quad (12)$$

where $\sigma(\cdot)$ is the softmax function. Through the 2-D convolutional decoder, we get the predicted segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$.

During training, we only use the one-shot setting, and its loss function $\mathcal{L}$ is given as follows:

$$\mathcal{L} = \mathrm{BCE}\left(\hat{\mathbf{Y}}_\mathbf{q}, \mathbf{Y}_\mathbf{q}\right) \quad (13)$$

where $\mathrm{BCE}(\cdot)$ is the binary cross-entropy loss between the predicted segmentation map $\hat{\mathbf{Y}}_\mathbf{q}$ and the ground truth $\mathbf{Y}_\mathbf{q}$. During inference, we simply generalize the one-shot setting to the K-shot setting through the voting mechanism [17].

IV. Experimental Results

In this section, we conduct experiments to verify the effectiveness of our proposed method. First, we briefly introduce the datasets and evaluation matrices used, as well as implementation details. Then, we report the performance through quantitative evaluation and compare our method with the SOAT methods. Next, we demonstrate the segmentation results through qualitative evaluation. Finally, a series of ablation experiments are conducted to illustrate the importance of each module and its contribution to performance.

A. Datasets

Two common datasets in remote sensing imagery are used to verify the performance of our proposed method.

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TABLE I
DETAILS OF iSAID-5^i. # REPRESENTS THE NUMBER OF IMAGES

Class ID	Fold	Class Name	#Training	#Test
C1	Fold 0	ship	4335	1592
C2		storage tank	961	280
C3		baseball diamond	542	240
C4		tennis court	2592	813
C5		basketball court	660	183
C6	Fold 1	ground track field	1478	547
C7		bridge	246	96
C8		large vehicle	3931	1190
C9		small vehicle	5302	1814
C10		helicopter	78	10
C11	Fold 2	swimming pool	243	74
C12		roundabout	219	54
C13		soccer ball field	2131	738
C14		plane	1890	1000
C15		harbor	3624	1432

TABLE II
DETAILS OF DLRSD-5^i. # REPRESENTS THE NUMBER OF IMAGES

Class ID	Fold	Class Name	#Image
C1	Fold 0	airplane	100
C2		bare soil	754
C3		buildings	713
C4		cars	897
C5		chaparral	115
C6	Fold 1	court	105
C7		dock	100
C8		field	103
C9		grass	977
C10		mobile home	101
C11	Fold 2	pavement	1331
C12		sand	291
C13		sea	101
C14		ship	103
C15		tanks	99

1) iSAID-5^i: The iSAID-5^i dataset is constructed from the iSAID [61] dataset by Yao et al. [23]. The iSAID dataset contains 2806 high-resolution images with a total of 655 451 finely annotated object instances in 15 semantic categories. Following the setting of Yao et al. [23], these 15 semantic categories are evenly divided into three folds, which are respectively used as unseen categories under the few-shot setting, as shown in Table I. In iSAID-5^i, the image size of all images is fixed at 256×256. Similar to Yao et al. [23] and Wang et al. [17], our proposed method is trained and evaluated using a threefold cross-validation method. Specifically, for Fold i ($i = 0, 1,$ or 2), we first train our proposed method based on the training images of the other two folds and then evaluate our proposed method on the test images of Fold i.

2) DLRSD-5^i: The DLRSD-5^i dataset is constructed from the DLRSD [62] dataset by Wang et al. [17]. The DLRSD dataset contains a total of 2100 RGB images in 21 semantic categories, and the image size of all images is fixed at 256×256. Following the setting of Wang et al. [17], 15 semantic categories from these categories are taken and divided equally into three folds, which are respectively used as unseen categories under the few-shot setting, as shown in Table II. Similar to iSAID-5^i, our method is trained and evaluated using a threefold cross-validation method. Note that in DLRSD-5^i, since the number of images per class is small and the training and testing classes do not overlap, all images within a class are no longer divided into training and validation subsets.

B. Evaluation Metric

Following common practice [17], [23], mean Intersection over Union (mIoU) is adopted as our evaluation metric. For each category, the IoU is calculated by the confusion matrix: $\text{IoU} = (\text{TP}/\text{TP} + \text{FN} + \text{FP})$, where TP, FN, and FP represent true positive, false negative, and false positive in the confusion matrix, respectively. The meaning of mIoU is the average of IoU over all categories in a fold: $\text{mIoU} = (1/C)\sum_{c=1}^{C} \text{IoU}_c$, where C is the number of categories in each fold and IoU_c is the IoU score of the cth category. Following the widely adopted method [17], [50], [51], [52], we evaluate and report performance under one- and five-shot settings.

C. Implementation Details

For the frozen SAM [27], we use the ViT-H architecture, which contains 32 layers of Transformer [63], each layer of multihead attention contains 16 heads, and the feature dimension is 1280. The size of the input image is unified to 1024×1024. Same as SAM [27], also no data augmentation is applied. During mask generation, the number of points to be sampled along one side of the image is 32 and the stability score threshold is 0.5. After masks are generated, the predicted masks are sorted according to the area size of the mask, and the first 128 masks are taken for subsequent processing, and if there are less than 128, empty masks are filled, i.e., $m = 128$. For example, if the number of predicted masks is 120, then eight empty masks are added so that the number of masks is 128.

In terms of the backbone, for a fair comparison, we use the common pretrained models on ImageNet [64] as the backbone, such as VGG16 [57], ResNet50 [58], and ResNet101 [58]. From these backbones, we extract three level features, i.e., $L = 3$. For the backbone of VGG16, we extract features after the last three max-pooling layers, respectively. For the backbones of ResNet50 and ResNet101, we extract features after the last three stages, respectively. In MGCD, $\bar{c}$ is set to 64.

The image size of both datasets is 256×256, and no data augmentation methods are used during our training process. For all experiments, we use SGD to optimize the proposed method with a weight decay of 5e-4 and a momentum of 0.9. For the iSAID-5^i dataset, our method is trained for 50 epochs, the initial learning rate of weights is 1e-2, and the initial learning rate of biases is 2e-2. For the DLRSD-5^i dataset, our method is trained for 100 epochs, the initial learning rate of weights is 1e-3, and the initial learning rate of biases is 2e-3. The backbone network is trained together with a 1000 times smaller learning rate. On both datasets, each training batch contains 16 support–query pairs. All VGG16- and ResNet50-based experiments are performed on a single NVIDIA GTX

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TABLE III

PERFORMANCE MEASURED WITH MIOU UNDER ONE- AND FIVE-SHOT SETTINGS ON THE iSAID-5^i DATASET. **BOLD** RESULTS INDICATE THE BEST PERFORMANCE, UNDERLINED RESULTS INDICATE THE SECOND BEST, RED RESULTS REPRESENT HOW MUCH PERFORMANCE IS IMPROVED, AND GREEN RESULTS REPRESENT HOW MUCH PERFORMANCE IS DEGRADED. THE RESULTS OF THE COMPARISON METHODS ARE FROM SCCNET [17]

Backbones	Methods	1-shot				5-shot				Learnable
		Fold 0	Fold 1	Fold 2	Mean	Fold 0	Fold 1	Fold 2	Mean	Parameters
VGG16	PANet [32]	17.43	11.43	15.95	14.94	17.70	14.58	20.70	17.66	14.7M
	PFENet [45]	16.68	15.30	27.87	19.95	18.46	18.39	28.81	21.89	10.4M
	PMMs [43]	20.87	16.07	24.65	20.53	23.31	16.61	27.43	22.45	25.8M
	CANet [44]	19.73	17.98	30.93	22.88	23.45	20.53	30.12	24.70	26.4M
	HSNet [20]	22.74	23.05	25.76	23.84	27.20	28.86	28.82	28.29	2.6M
	SDM [23]	29.24	20.80	34.73	28.26	36.33	27.98	42.39	35.57	25.8M
	SCCNet [17]	30.00	27.41	32.43	29.94	36.52	**31.40**	37.53	35.15	5.2M
	Ours	**38.24**	**28.61**	**40.21**	**35.69**	**43.81**	31.07	**48.31**	**41.06**	**0.7M**
		+8.24	+1.20	+5.48	+5.75	+7.29	-0.33	+5.92	+5.49	-
ResNet50	PANet [32]	12.36	9.11	12.05	11.17	13.82	12.4	19.12	15.11	23.5M
	PFENet [45]	18.75	17.24	22.09	19.36	19.57	18.43	26.14	21.38	10.8M
	CANet [44]	18.80	15.62	25.79	20.07	23.86	18.54	32.00	24.80	20.2M
	PMMs [43]	19.02	18.51	28.42	21.98	20.89	20.87	31.23	24.33	19.6M
	SDM [23]	34.29	22.25	35.62	30.72	39.88	30.59	45.70	38.72	19.6M
	HSNet [20]	30.76	24.35	38.20	31.10	38.08	30.56	45.28	37.79	2.6M
	SCCNet [17]	36.21	27.42	43.37	35.67	42.58	30.30	50.26	41.05	5.2M
	Ours	**42.77**	**30.59**	**46.39**	**39.92**	**49.14**	**32.58**	**52.75**	**44.82**	**0.7M**
		+6.56	+3.17	+3.02	+4.25	+6.56	+1.99	+2.49	+3.77	-
ResNet101	HSNet [20]	34.91	26.51	40.84	34.09	41.71	31.08	48.54	40.44	2.6M
	SCCNet [17]	37.65	29.19	42.99	36.13	41.87	32.12	49.63	41.20	5.2M
	Ours	**45.21**	**30.23**	**47.13**	**40.86**	**50.58**	**31.38**	**54.02**	**45.33**	**0.7M**
		+7.56	+1.04	+4.14	+4.73	+8.71	-0.74	+4.39	+4.13	-

1080Ti GPU, and all ResNet101-based experiments are performed on two NVIDIA GTX 1080Ti GPUs. All experiments are implemented based on PyTorch [65].

D. Comparison With the SOAT

To evaluate the effectiveness of the proposed method, we report results with different backbone networks and few-shot settings on the iSAID-5^i and DLRSD-5^i datasets and extensively compare them with the SOAT methods.

1) Comparison on the iSAID-5^i dataset: The experimental results of our method and the comparison results with advanced methods on the iSAID-5^i dataset are shown in Table III. In Table III, all methods report results measured in the metric of mIoU under one- and five-shot settings. The results of our method are shown in rows "Ours." Among all the methods, it can be seen from Table III that in the one-shot setting, our method performs best in all three folds, and the mean mIoU is increased by 5.75 based on VGG16, 4.25 based on ResNet50, and 4.73 based on ResNet101 compared to SCCNet [17]. Due to the inability to exploit the inherent characteristics of the iSAID-5^i dataset, traditional methods such as PANet [32], PFENet [45], PMMs [43], and CANet [44] are difficult to perform effectively on this dataset. Our FBC module models the foreground and background correlation separately, which alleviates the low foreground–background contrast characteristics in the iSAID-5^i dataset and greatly helps improve performance, which is demonstrated in the ablation experiments in Section IV-E1. It also can be seen that in the five-shot setting, our method based on VGG16 and ResNet101 performs best in two folds and second best in Fold 1, and the mean mIoU of Fold 1 is increased by 5.49 based on VGG16 compared with SDM [23] and 4.13 based on ResNet101 compared with SCCNet [17]. The reason why the best performance is not achieved in Fold 1 may be that our method does not achieve improvement on the category of "small vehicle" with very small scales. In this category, we only achieve an mIoU of 13.19, which is much lower than other categories and lower than the comparison methods, SCCNet [17] and PFENet [45]. In the iSAID-5^i dataset, the category "small vehicle" has a smaller resolution and lower clarity, resulting in poor segmentation results.

It is worth noting that the parameters of our proposed modules are only 0.7M, which is much less than other methods. Overall, our method achieves the best performance among all three backbones under one- and five-shot settings with less than 1M additional parameters. Compared with other methods, our method achieves the best results with fewer parameters largely because we utilize high-quality oversegmented masks to guide the network to enhance the consistency of features within the masks, which is demonstrated in the ablation experiments in Section IV-E1. In addition to the above quantitative evaluation, we show some predicted segmentation maps under the one-shot setting on iSAID-5^i, as shown in Fig. 6. It can be seen that our method can achieve better segmentation in a variety of scenarios. By utilizing high-quality masks, the segmentation results of our method have more realistic and complete boundaries than SCCNet [17].

2) Comparison on the DLRSD-5^i dataset: The experimental results of our proposed method and the comparison results with the SOAT methods on DLRSD-5^i are shown in Table IV.

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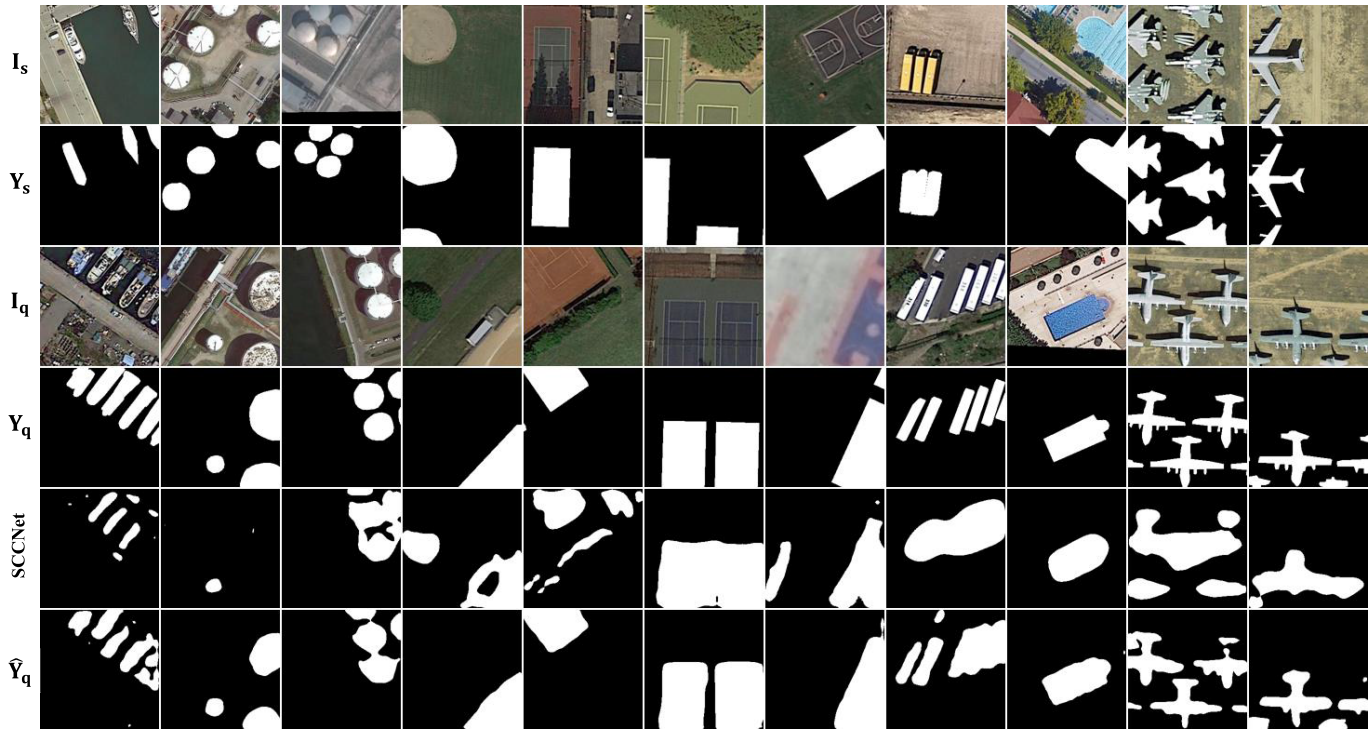


Fig. 6. Predicted segmentation maps under the one-shot setting on the iSAID-5^i dataset. $\mathbf{I_s}$ and $\mathbf{I_q}$ are the support and query images, $\mathbf{Y_s}$ and $\mathbf{Y_q}$ are the segmentation map of the support and query images, SCCNet represents the result of SCCNet [17], and $\hat{\mathbf{Y}}_\mathbf{q}$ is our predicted segmentation map.

TABLE IV

PERFORMANCE MEASURED WITH MIOU UNDER ONE- AND FIVE-SHOT SETTINGS ON THE DLRSD-5^i DATASET. **BOLD** RESULTS INDICATE THE BEST PERFORMANCE, UNDERLINED RESULTS INDICATE THE SECOND BEST, AND RED RESULTS REPRESENT HOW MUCH PERFORMANCE IS IMPROVED. THE RESULTS OF THE COMPARISON METHODS ARE FROM SCCNET [17]

Backbones	Methods	1-shot				5-shot				Learnable Parameters
		Fold 0	Fold 1	Fold 2	Mean	Fold 0	Fold 1	Fold 2	Mean	
ResNet50	DCAMA [66]	12.68	13.01	10.60	12.10	14.37	15.67	15.24	15.09	-
	PFENet [45]	21.98	20.67	11.17	17.94	24.12	24.76	16.01	21.63	10.8M
	SDM [23]	20.11	30.84	27.87	26.27	26.03	41.74	33.55	33.77	19.6M
	MLC [67]	26.38	36.05	34.65	32.36	31.22	40.33	43.03	38.19	-
	HSNet [20]	22.00	47.20	34.73	34.64	27.46	52.32	46.23	42.00	2.6M
	PCFNet [68]	**31.09**	37.52	41.89	36.83	**36.67**	42.70	**54.20**	44.52	-
	SCCNet [17]	25.34	48.97	39.73	37.37	30.22	52.40	47.15	43.26	5.2M
	Ours	30.37	**50.57**	**43.01**	**41.32**	33.64	**54.84**	52.34	**46.94**	**0.7M**
		-0.72	+1.60	+1.12	+3.95	-3.03	+2.44	-1.86	+2.42	-

In Table IV, all methods report results measured in the metric of mIoU under one- and five-shot settings with ResNet50 as the backbone. The results of our method are shown in rows "Ours." It can be seen from Table IV that under the both one- and five-shot settings, our method performs best on the average of the three folds. In particular, the mean mIoU is increased by 3.95 under the one-shot setting and 2.42 under the five-shot setting compared to SCCNet [17] and PCFNet [68]. These results show that by integrating well-designed MGFE, FBC, and MGCD modules, the overall performance of the model can be significantly improved, which is also demonstrated in Section IV-E1. It is worth noting that the parameters of our proposed modules are only 0.7M, which is much less than other methods. Overall, our method achieves the best performance with ResNet50 as the backbone under one- and five-shot settings with less than 1M additional learnable parameters. Compared with other methods, our method achieves the best results with fewer parameters by leveraging high-quality masks to guide the network to enhance the intramask consistency of the extracted features.

E. Ablation Experiment

To analyze our proposed modules and their hyperparameters, we conduct a series of experiments using ResNet50 as the backbone on the iSAID-5^i dataset under the one-shot setting.

1) Effect Analysis of Each Module: In our method, the novelty mainly lies in the MGFE and FBC modules, where MGFE is applied in the support image branch, the query image branch, and the MGCD module, marked as MGFE_S, MGFE_Q, and MGFE_MGCD, respectively. To explore the role

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TABLE V

EFFECT ANALYSIS OF EACH MODULE. RESULTS MEASURED USING MIOU ON ISAID-5^i. WITHOUT FBC MEANS LEARNING ONLY THE FOREGROUND CORRELATION, MGFE_S MEANS MGFE IN THE SUPPORT IMAGE BRANCH, MGFE_Q MEANS MGFE IN THE QUERY IMAGE BRANCH, MGFE_MGCD MEANS MGFE IN THE MGCD MODULE, AND RED RESULTS MEAN INDICATE HOW MUCH PERFORMANCE IMPROVED COMPARED TO THE BASELINE

FBC	MGFE_S	MGFE_Q	MGFE_MGCD	1-shot				5-shot			
				Fold 0	Fold 1	Fold 2	Mean	Fold 0	Fold 1	Fold 2	Mean
×	×	×	×	34.25	26.04	37.13	32.47	40.82	28.75	44.61	38.06
✓	×	×	×	37.29	28.50	39.22	35.00 +2.53	43.36	30.32	47.14	40.27 +2.21
×	✓	✓	✓	37.65	28.20	41.59	35.81 +3.34	44.11	30.08	48.52	40.90 +2.84
✓	✓	×	×	38.67	27.82	41.15	35.88 +3.41	44.98	30.67	48.28	41.31 +3.25
✓	×	✓	×	40.45	28.18	41.48	36.70 +4.23	46.03	31.26	49.73	42.34 +4.28
✓	×	×	✓	41.84	29.91	44.84	38.86 +6.39	47.85	32.01	51.07	43.64 +5.58
✓	✓	✓	×	40.16	28.86	43.18	37.40 +4.93	46.68	31.52	50.08	42.76 +4.70
✓	✓	✓	✓	42.77	30.59	46.39	39.92 +7.45	49.14	32.58	52.75	44.82 +6.76

TABLE VI

ABLATION STUDY OF HYPERPARAMETER L. RESULTS MEASURED USING MIOU UNDER THE ONE-SHOT SETTING ON THE ISAID-5^i DATASET. RED RESULTS INDICATE HOW MUCH PERFORMANCE IS IMPROVED

L	Fold 0	Fold 1	Fold 2	Mean
1	29.24	23.74	32.59	28.52
2	36.82	26.94	41.14	34.97 +6.45
3	42.77	30.59	46.39	39.92 +4.95
4	43.23	31.11	46.47	40.27 +0.35

TABLE VII

ABLATION STUDY OF HYPERPARAMETER m. RESULTS MEASURED USING MIOU UNDER THE ONE-SHOT SETTING ON THE ISAID-5^i DATASET. **BOLD** RESULTS MEAN THE BEST RESULTS

m	Fold 0	Fold 1	Fold 2	Mean
×	37.29	28.50	39.22	35.00
8	39.86	28.01	43.69	37.19
16	41.71	29.14	45.99	38.95
32	42.67	30.44	46.32	39.81
64	42.74	**30.63**	46.15	39.84
128	**42.77**	30.59	**46.39**	**39.92**
256	42.52	30.53	46.23	39.76

and contribution of each module to the performance, we conduct a set of ablation experiments on the iSAID-5^i dataset with ResNet50 as the backbone. We measure the mIoU metric under one- and five-shot settings, and the experimental results are shown in Table V. It is clear that our proposed MGFE and FBC can lead to good results. Specifically, first, the model with only FBC improves by 2.53 under the one-shot setting and 2.21 under the five-shot setting compared with the baseline (the first row). This shows that adding background correlation based on foreground correlation can extract better features to distinguish foreground and background, thereby reducing the impact of low foreground–background contrast. Furthermore, models with MGFE achieve significant improvements in the mIoU metric, such as the third and last rows in Table V. This indicates that improving the consistency within feature masks through MGFE helps improve the accuracy of segmentation results. In particular, the combination of FBC and MGFE brings a huge improvement, which implies that background-related masks have the effect of inhibiting the 2-D convolutional decoder from segmenting the background. In addition, we explore the effect of each MGFE module respectively in rows 4, 5, and 6. The results show that the MGFE module in MGCD has the largest improvement. This also shows that the oversegmented masks play the most effective guiding role when it is close to the final 2-D convolutional decoder. Overall, the above experimental results demonstrate that each novel module in our method can bring performance improvements.

2) Ablation Study of Hyperparameter L in 4-D Convolutional Pyramid Encoder: The hyperparameter L is the level of the feature pyramid in a 4-D convolutional pyramid encoder. Generally, the more levels there are, the higher the accuracy of the predicted segmentation maps and the greater the amount of computation. To analyze the impact of hyperparameter L on performance, we conduct a set of ablation experiments under the one-shot setting on the iSAID-5^i dataset. The results are shown in Table VI. We can see that the hyperparameter L has a crucial impact on performance. Specifically, when $L = 4$, the best performance is achieved under the one-shot setting, which is 11.75 higher than when $L = 1$. It can be concluded that when $L = 4$, the mIoU metric is only improved by 0.35 compared with $L = 3$, and when $L = 3$, the mIoU metric is improved by 4.97 compared with $L = 2$. The improvement in mIoU brought by $L = 4$ comes at the expense of increased training time and memory usage. In addition, the comparison methods, HSNet [20] and SCCNet [17], directly adopt the setting of $L = 3$. In order to reduce training time, inference time, and memory usage, and for the sake of fairness, we also set L to 3 in our experiments although $L = 4$ is a slight improvement.

3) Ablation Study of Hyperparameter m: The hyperparameter m represents the number of masks used in MGFE. When the value of m is small, the pooling index $\mathbb{I}$ appears coarse-grained, and when the value of m is large, the pooling index $\mathbb{I}$ appears fine-grained. Therefore, the hyperparameter m affects the effect of feature enhancement. To analyze the hyperparameter m, we conduct a parametric analysis experiment under the one-shot setting on the iSAID-5^i dataset. The results are shown in Table VII, where × means that there is no MGFE module. It can be seen that when the hyperparameter m is 64 or 128,

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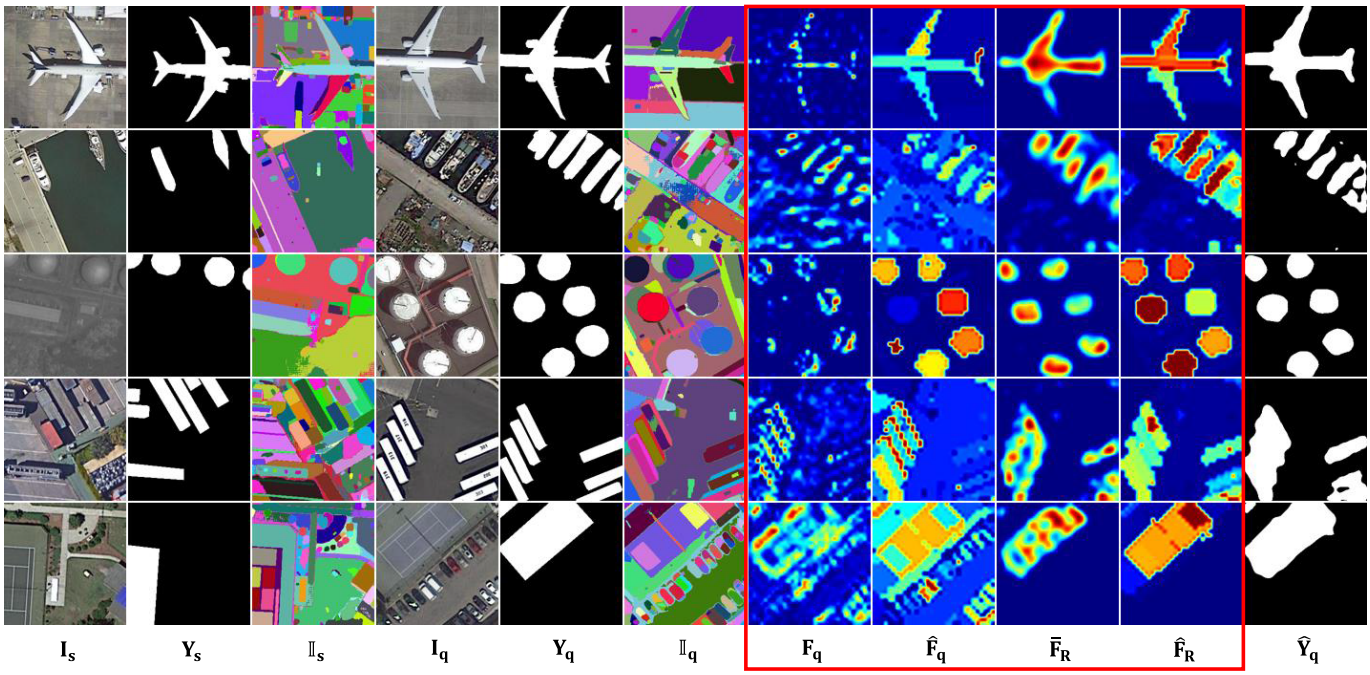


Fig. 7. Feature enhancement effect of our proposed MGFE under the one-shot setting on the iSAID-5^i dataset. $\mathbf{I_s}$, $\mathbf{Y_s}$, and $\mathbb{I}_\mathbf{s}$ are the support image and its segmentation map and pooling index, respectively. $\mathbf{I_q}$, $\mathbf{Y_q}$, and $\mathbb{I}_\mathbf{q}$ are the query image and its segmentation map and pooling index, respectively. $\mathbf{F_q}$, $\hat{\mathbf{F}}_\mathbf{q}$, $\bar{\mathbf{F}}_\mathbf{R}$, and $\hat{\mathbf{F}}_\mathbf{R}$ are the query feature of extracted from the backbone, the enhanced query feature by the MGFE module, the correlation feature, and the enhanced correlation feature by the MGFE module in the MGCD module, respectively. $\hat{\mathbf{Y}}_\mathbf{q}$ is the predicted segmentation map of the query image.

TABLE VIII

ACCURACIES FOR EACH CATEGORY. RESULTS MEASURED WITH MIOU UNDER THE ONE-SHOT SETTING. **BOLD** RESULTS INDICATE THE BEST PERFORMANCE, UNDERLINED RESULTS INDICATE THE SECOND BEST, RED REPRESENTS HOW MUCH PERFORMANCE IS IMPROVED, AND GREEN REPRESENTS HOW MUCH PERFORMANCE IS DEGRADED. THE RESULTS OF THE COMPARISON METHODS ARE FROM SCCNET [17] AND SDM [23]

Datasets	Methods	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15
iSAID-5^i	PANet [32]	7.27	9.99	14.49	13.20	16.85	17.40	10.23	7.37	6.21	4.33	8.35	12.38	21.28	8.52	9.75
	CANet [44]	7.25	27.25	43.32	6.17	9.99	6.16	17.81	28.58	8.84	16.73	17.73	56.97	13.82	27.41	13.06
	PMMs [43]	12.93	17.92	41.97	7.43	14.87	13.24	32.67	26.08	4.17	16.37	12.26	61.49	38.37	17.92	12.02
	PFENet [45]	2.98	6.30	33.29	34.49	16.71	8.68	8.67	34.62	**18.19**	16.08	28.58	24.47	7.37	39.76	10.31
	SDM [23]	**37.66**	34.37	34.45	39.81	25.14	16.77	34.53	30.50	12.42	17.02	20.69	56.83	42.80	40.52	17.26
	HSNet [20]	18.93	30.01	37.60	45.33	21.95	25.11	37.17	27.43	11.03	21.01	32.22	50.07	**54.27**	37.98	16.46
	SCCNet [17]	26.76	43.42	40.27	46.74	22.10	27.37	36.75	32.94	14.53	23.89	**46.85**	55.06	45.77	48.02	23.30
	Ours	33.75	**46.60**	**49.52**	**55.50**	**28.50**	**32.14**	**45.51**	**35.79**	13.19	**26.34**	45.73	**61.66**	46.11	**53.89**	**24.56**
		-3.91	+3.18	+6.20	+8.76	+3.36	+4.77	+8.34	+1.17	-5.00	+2.45	-1.12	+0.17	-8.16	+5.87	+1.26
DLRSD-5^i	SDM [23]	5.51	22.74	29.00	3.83	39.49	5.30	19.97	84.13	8.94	35.90	11.96	31.99	49.03	38.79	7.57
	HSNet [20]	23.63	18.39	21.41	8.55	38.02	**63.45**	24.56	**96.49**	18.33	33.20	20.88	24.66	57.13	35.00	35.94
	SCCNet [17]	23.58	25.32	26.99	**10.45**	40.37	53.27	**25.49**	96.29	29.10	40.68	**30.09**	**24.80**	60.07	**46.72**	37.00
	Ours	**37.21**	**32.22**	**33.45**	7.60	**41.35**	56.99	21.36	91.80	**35.22**	**47.51**	26.16	23.04	**68.78**	41.98	**55.08**
		+13.58	+6.90	+4.45	-4.73	+0.98	-6.46	-4.13	-4.69	+6.12	+6.83	-3.93	-1.76	+8.71	-4.74	+18.08

the best performance is achieved under the one-shot setting. After taking all folds into account, we set m to 128 in all other experiments. In addition, it can also be seen from Table VII that when m is less than 32, the mIOU metric increases significantly, and when m is greater than 32, the mIOU metric increases slowly, and even performance drops in some folds. Therefore, it does not mean that the finer the granularity of the mask, the better the segmentation performance of our method. In our setting, the number of masks obtained by SAM for an image is usually within 150 and has many masks with small mask areas. Therefore, when $m = 256$, some of the masks are empty masks added, and some are masks with small mask areas, so the performance of our method decreases, as shown in the last row of Table VII.

F. Analysis and Visualization

In this section, we further analyze our method from the perspective of visualization and performance of each category.

1) Performance Analysis for Each Category: As shown in Table VIII, the one-shot mIoU of each category on the iSAID-5^i and DLRSD-5^i datasets with ResNet50 as the backbone is reported. As can be seen from Table VIII, our method achieves the best mIoU metrics in most categories on both datasets. For example, on the iSAID-5^i dataset, the mIoU of the category "baseball diamond" (C3) is 49.52, which is 6.20 higher than the previous best result, and the mIoU of the category "tennis court" (C4) is 55.50, which is 8.76 higher than the previous best result. In our method, the mIoU for the category "small

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vehicle" (C9) is only 13.19, which achieves poor results compared with other categories. C9 is the category of "small vehicle" with very small scales. Since only a three-level feature pyramid is used and the feature size is 32×32, the performance of our method is poor for such small objects as "small vehicle". Each image in the iSAID-5^i dataset is of the size in the range from 800×800 to $20\,000 \times 20\,000$ pixels, while the size of each image in the DLRSD-5^i dataset is 256×256 pixels. As we pointed out in the implementation details, when extracting masks via SAM, all images are unified to 1024×1024 pixels. Since the resolution of the images in the DLRSD-5^i dataset is lower, the quality of the obtained masks is worse than in the iSAID-5^i dataset. This results in some categories having lower IoU on the DLRSD-5^i dataset, such as "cars" (C4), "court" (C6), "dock" (C7), "field" (C8), and "ship" (C14). Based on this situation, we will make further improvements in future work.

2) Visualization of MGFE: To further illustrate that our method has intramask feature consistency, as shown in Fig. 7, we visualize multiple sets of features before and after the MGFE module on the iSAID-5^i datasets. It is obvious from column $\mathbf{F_q}$ to column $\hat{\mathbf{F}}_\mathbf{q}$ in the red box that the features are enhanced according to the corresponding pooling index. It also can be seen from column $\bar{\mathbf{F}}_\mathbf{R}$ to column $\hat{\mathbf{F}}_\mathbf{R}$ in the red box that the enhanced correlation features are very similar to the ground-truth segmentation maps. In Fig. 7, $\bar{\mathbf{F}}_\mathbf{R}$ and $\hat{\mathbf{F}}_\mathbf{R}$ are bilinearly interpolated from 32×32 to 128×128. Since $\hat{\mathbf{F}}_\mathbf{R}$ has been guided by the MGFE module, it presents a border similar to the mask, but its border has a jagged effect. $\hat{\mathbf{F}}_\mathbf{R}$ is the feature before the MGFE module, so there is no such effect. In our method, $\bar{\mathbf{F}}_\mathbf{R}$ and $\hat{\mathbf{F}}_\mathbf{R}$ are fed into the 2-D convolutional decoder to get the final result $\hat{\mathbf{Y}}_\mathbf{q}$. As can be seen from $\hat{\mathbf{Y}}_\mathbf{q}$ in Fig. 7, this effect disappears after passing through the 2-D convolutional decoder. In our MGFE module, as shown in (4), the MAP operation averages the features within the same mask, and as shown in (5), the RMAP operation backfills the averaged features into the corresponding masks. Through MAP and RMAP, our MGFE module successfully achieves intramask feature consistency, thereby improving the performance of few-shot segmentation in remote sensing imagery.

3) Time Cost in the Evaluation: Our method utilizes SAM to generate oversegmented masks, which consumes most of the time. For the one-shot setting, it is necessary to obtain the oversegmented masks of a support image and a query image through SAM. On a single NVIDIA GTX 1080Ti GPU, it takes 1.8 s to process these two images of size 256×256 and then 0.09 s to get the final predicted segmentation map. Therefore, a one-shot task takes 1.89 s to evaluate. In order to speed up the training and evaluation process, in our experiments, we preobtain and save the oversegmented masks of all images through SAM.

V. Conclusion

In this work, we propose a novel MGCL method for few-shot segmentation in remote sensing imagery. Particularly, by using oversegmented masks to guide how to enhance features, the MGFE module makes the enhanced features have the consistency of features within the mask. Then, by learning foreground correlation and background correlation separately, the FBC module enhances the representation of background correlation and reduces the impact of low foreground–background contrast in remote sensing images. Furthermore, in the decoding of correlation learning, the MGCD module guides the decoder to pay attention to the consistency within the mask and learn how to segment complete objects, thereby improving the accuracy of segmentation. Overall, our proposed method achieves the best segmentation results on two common datasets and establishes a large performance gap with previous methods. The visualization of features before and after the MGFE module further concretely demonstrates the motivation and advantages of our proposed method. In the future, we will explore how to better utilize oversegmented masks.

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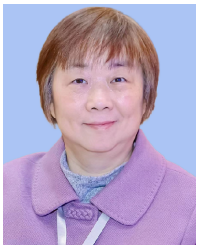
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